General Theory of Doubly Periodic Arrays over an Arbitrary Finite Field and Its Applications

SHOJIRO SAKATA

Abstract—**A general theory of doubly periodic (DP) arrays over an arbitrary finite field GF(q) is presented. First the basic properties of DP arrays are examined. Next modules of linear recurring (LR) arrays are defined and their algebraic properties discussed in connection with ideals in an extension ring $\tilde{R}$ of the ring R of bivariate polynomials with coefficients in GF(q). A finite $\tilde{R}$-module of DP arrays is shown to coincide with the $\tilde{R}$-module of LR arrays defined by a zero-dimensional ideal in $\tilde{R}$. Equivalence relations between DP arrays are explored, i.e., rearrangements of arrays by means of unimodular transformations. Decimation and interleaving of arrays are defined in a two-dimensional sense. The general theory is followed by application to irreducible LR arrays. Among irreducible arrays, M-arrays are a two-dimensional analog of M-sequences and may be constructed from M-sequences by means of unimodular transformations. The results of this paper are also important in studying properties of Abelian codes.**

I. Introduction

DOUBLY periodic (DP) arrays over an arbitrary finite field GF(q) are a two-dimensional generalization of periodic sequences over GF(q). Linear recurring sequences which are generated by a linear shift register with feedback are eventually periodic sequences. A linear recurring sequence is determined, given a set of initial values, by a linear recurring relation or by a homogeneous difference equation which can be implemented with a linear feedback shift register.

Our investigation mostly concerns a general property of purely periodic arrays, i.e., the general solutions for systems of homogeneous linear partial difference equations. It has not been mentioned by other authors on linear recurring (LR) arrays or planes [1]–[4] that the "period" of a DP array is not completely characterized by the partial periods $\mathrm{per}_x\ (u)$ and $\mathrm{per}_y\ (u)$. We should also compare DP arrays with elliptic functions, which are doubly periodic on the complex plane [8].

This approach is especially important in studying two-dimensional cyclic codes in connection with LR arrays. It is similarly possible to investigate multidimensional periodic arrays and multidimensional cyclic codes. Multidimensional cyclic codes are also called Abelian codes, which are a natural generalization of cyclic codes. An Abelian code is defined to be an ideal in an Abelian group ring. Although several authors [9]–[13] have observed that many properties of cyclic codes carry over to Abelian codes, some intrinsic properties remain for further investigations.

A general theory of two-dimensional cyclic codes has been proposed by Ikai *et al.* [14]–[16], where a "quasi-division" algorithm in the bivariate polynomial ring is introduced for the purpose of determining the positions of check symbols as well as generator polynomials. This technique has also been applied to develop a theory of two-dimensional LR arrays [3]. It will be shown in the present paper, however, that a more general approach is possible without invoking such a special device. Our method is based only on the fundamental concepts and theorems of ideals and modules and is therefore of wide applicability. It will also be important in devising more efficient encoding and decoding methods for Abelian group codes.

Irreducible LR arrays are DP arrays of which the characteristic ideal $\mathfrak{P}$ is prime. They have the simplest algebraic structure and can be treated by a special device: the algebraic set $V(\mathfrak{P})$, i.e., the zeros belonging to the ideal. M-arrays are the most important irreducible LR arrays. They are two-dimensional analogs of M-sequences or maximum length sequences. $\gamma\beta$-arrays have been introduced in [2] and have what is called the maximum area property.

In the present paper a general treatment of DP and LR arrays will be presented. The basic mathematical properties are studied using the theory of modules over the ring $\tilde{R}$ which is isomorphic to an extension ring of the bivariate polynomial ring R. Correspondence relations between DP arrays are clarified by means of the automorphisms of the ring $\tilde{R}$ and $\tilde{R}$-isomorphisms between $\tilde{R}$-modules.

The outline of the paper is as follows. Section II is devoted to preliminaries on DP and LR arrays. Section III describes the fundamental theorem which clarifies the relation between DP and LR arrays. The algebraic properties of LR arrays are explored in connection with their characteristic ideals. The independent point set, i.e., the positions of the information symbols in the terminology of algebraic coding theory, is determined in general. Section IV describes various correspondences between DP arrays

Manuscript received August 1, 1977; revised April 18, 1978. Part of the results described in this paper were presented at conferences of the Technical Group on Automata and Languages, Institute of Electronics and Communications Engineers of Japan, December 1976 and March 1977, [5], [6].

The author is with the Department of Information Science, Faculty of Engineering, Sagami Institute of Technology, Fujisawa 251, Japan.

0018-9448/78/1100-0719$00.75 ©1978 IEEE

Authorized licensed use limited to: Lakehead University. Downloaded on September 25,2026 at 17:14:10 UTC from IEEE Xplore. Restrictions apply.

determined by rearrangement, decimation, and interleaving, which are associated with automorphisms and endomorphisms of $\tilde{R}$. In Section V the general treatment of DP and LR arrays will be applied to irreducible LR arrays. In a two-dimensionally proper sense we will define M-arrays, which are shown to coincide with $\gamma\beta$-arrays and to be constructed from M-sequences by rearrangements.

In this paper we make free use of concepts and theorems from commutative ring theory (e.g., [17], [18]) and matrix theory.

II. Preliminaries

The two-dimensional integral lattice is the totality of lattice points which have coordinates $(i,j)\in Z^2(\triangleq Z\times Z)$, where Z is the ring of integers. A *two-dimensional array* $u=(u_{i,j})$ is an arrangement of elements $u_{i,j}$ of $F\triangleq \mathrm{GF}(q)$ on the lattice points, where q is a power of a prime p. The value $u_{i,j}$ of an array u on the point $\boldsymbol{i}=(i,j)\in Z^2$ is said to be the (i,j)-component of u and is denoted alternatively by $(u)_{i,j}$ or by $u(\boldsymbol{i})$. A two-dimensional array may be identified with a double sequence $\{u_{i,j}\}$ with values in F or an infinite matrix $\boldsymbol{U}$ over F: $\boldsymbol{U}=\|u_{i,j}\|$.

We investigate *periodic* arrays in the following sense. They are a discrete analog of periodic functions with two indeterminates [8].

Definition 1: If an array u satisfies the relation $u(\boldsymbol{i})=u(\boldsymbol{i}+\boldsymbol{l})$, for some $\boldsymbol{l}\in Z^2$ and all $\boldsymbol{i}\in Z^2$, the integral vector $\boldsymbol{l}$ is called a *period* (P.) *vector* of u. An array with a P. vector other than the zero vector $\boldsymbol{0}$ is called a *periodic* array.

The set P of all P. vectors of u forms an additive group contained in Z^2. If a basis of this additive group exists, its members are called *fundamental period* (FP) *vectors* of u. An array with two FP vectors is called a *doubly periodic* array. (See Fig. 1.)

Let $\boldsymbol{l}_1,\boldsymbol{l}_2$ be the FP vectors of a DP array. For a given vector $\boldsymbol{l}\in Z^2$, the parallelogram with vertices $\boldsymbol{l}$, $\boldsymbol{l}+\boldsymbol{l}_1$, $\boldsymbol{l}+\boldsymbol{l}_2$, $\boldsymbol{l}+\boldsymbol{l}_1+\boldsymbol{l}_2$ (or the set $\boldsymbol{l}_1\times\boldsymbol{l}_2$ of lattice points expressed as $\boldsymbol{l}+\omega_1\boldsymbol{l}_1+\omega_2\boldsymbol{l}_2$, for rational numbers ω_1,ω_2 such that $0\leqslant\omega_1,\omega_2<1$) is called the *fundamental period parallelogram*. The two-dimensional plane (or integral alttice) is covered with a network of congruent parallelograms, called *period parallelograms*, obtained by translating the FP parallelogram through $m\boldsymbol{l}_1+n\boldsymbol{l}_2$, $m,n\in Z$.

For a basis $\boldsymbol{l}_1,\boldsymbol{l}_2$ of the additive group P, another basis $\boldsymbol{m}_1,\boldsymbol{m}_2$ is obtained from any unimodular matrix $\boldsymbol{S}$, i.e., an integral matrix with the determinant $\det(\boldsymbol{S})=1$ or -1, as follows:

$$\begin{Vmatrix}\boldsymbol{m}_1\\ \boldsymbol{m}_2\end{Vmatrix}=\boldsymbol{S}\begin{Vmatrix}\boldsymbol{l}_1\\ \boldsymbol{l}_2\end{Vmatrix}. \tag{1}$$

Example 1: Consider a DP array v over $\mathrm{GF}(2)=\{0,1\}$ with a FP parallelogram $\boldsymbol{l}_1\times\boldsymbol{l}_2$, where $\boldsymbol{l}_1=(4,2)$, $\boldsymbol{l}_2=(2,4)$. For example, v also has a FP parallelogram $\boldsymbol{m}_1\times\boldsymbol{m}_2$, where $\boldsymbol{m}_1=(4,2)$, $\boldsymbol{m}_2=(6,6)$. They are shown in Fig. 2.

The sum $u+v$ of two arrays u,v and scalar multiplication cu of an array u by a constant are defined in the usual way. Then the set W of all DP arrays forms an F-module, i.e., a linear space over the field F. The identity element of the module is the zero array (0). For two submodules U,V of W, $U+V$ is the submodule of W which is composed of the arrays of the form $u+v$ for $u\in U$ and $v\in V$.

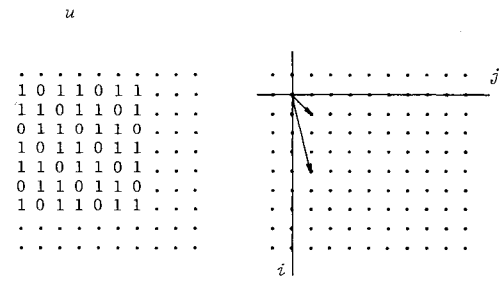


Fig. 1. Example of DP array over GF (2).

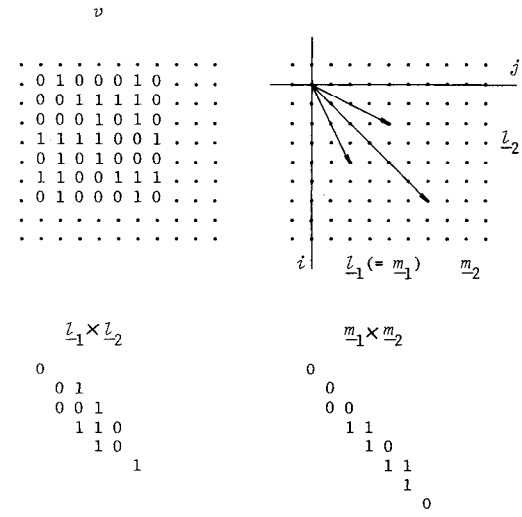


Fig. 2. Example of DP array over GF (2) and its FP parallelograms.

For some vector $\boldsymbol{k}=(k,l)\in Z^2$ and an array u, the array u' such that $u'(\boldsymbol{i})=u(\boldsymbol{i}+\boldsymbol{k})$, $\boldsymbol{i}\in Z^2$, is said to be the *translate* of u along $\boldsymbol{k}$ and is denoted by ${}_{\boldsymbol{k}}u$ or ${}_{k,l}u$. There are a finite number of different translates of a DP array u. Let the *cycle* of u denote the set of all different translates of u.

Definition 2: The cardinality of the cycle of a DP array u is called the *period* of u and is denoted by per (u).

per (u) is identical with the number of lattice points contained in a FP parallelogram $\boldsymbol{l}_1\times\boldsymbol{l}_2$ of u and is also equal to the absolute value of

$$\det\left(\begin{Vmatrix}\boldsymbol{l}_1\\ \boldsymbol{l}_2\end{Vmatrix}\right).$$

A DP array has a P. vector with an arbitrary direction $\boldsymbol{l}\in Z^2$. The length of a FP vector with the direction of $\boldsymbol{i}_x\triangleq(1,0)$ is said to be the *partial x-period* and is denoted by $\mathrm{per}_x(u)$. The *partial y-period* $\mathrm{per}_y(u)$ is defined in the same way. If $\mathrm{per}_x(u)$ or $\mathrm{per}_y(u)=1$, the array is the perpendicular repetition of a (simple) periodic sequence. This is the simplest case of double periodicity.

Authorized licensed use limited to: Lakehead University. Downloaded on September 25,2026 at 17:14:10 UTC from IEEE Xplore. Restrictions apply.

A DP array satisfies the trivial linear recurring relations $u_{i+r,j}=u_{i,j}$, $u_{i,j+s}=u_{i,j}$, $(i,j)\in Z^2$, for integers r, s such that $\mathrm{per}_x(u)|r$, $\mathrm{per}_y(u)|s$.[1] In order to deal with LR relations in general, we introduce polynomial-like operators $f(x,y)=\sum_{(I,J)\in\Gamma(f)}f_{I,J}x^Iy^J$, where $\Gamma(f)$ is a finite subset of Z^2 and $f_{I,J}$ are elements of F. By $f(x,y)u$ we mean the array v such that $v_{i,j}=\sum_{(I,J)\in\Gamma(f)}f_{I,J}u_{i+I,j+J}$, $(i,j)\in Z^2$. In particular, $x^ku=_{k,0}u$ and $y^lu=_{0,l}u$ are translates of u. These operators form a ring, which is isomorphic to the ring $\tilde{R}\triangleq\{x^iy^jf(x,y)|i,j\in Z,\ f(x,y)\in R\}$, where R is the ring of bivariate polynomials with coefficients in F. From now on we do not distinguish between the ring of operators and $\tilde{R}$.

Definition 3: For an ideal $\mathfrak{A}$ in the ring $\tilde{R}$, let $G(\mathfrak{A})$ denote the set of all arrays u such that, for any element $f(x,y)$ of $\mathfrak{A}$, $f(x,y)u=(0)$.

$G(\mathfrak{A})$ is an $\tilde{R}$-module, i.e., a module which is closed under any operations of $\tilde{R}$.

Any ideal $\mathfrak{A}$ in $\tilde{R}$ has a finite basis $\mathfrak{A}=(f^{(1)},\cdots,f^{(M)})$. (See Appendix A.) Clearly $G(\mathfrak{A})$ coincides with the set of linear recurring arrays u which satisfy the LR relations $f^{(k)}(x,y)u=(0)$, $1\leqslant k\leqslant M$, or equivalently, for $f^{(k)}(x,y)=\sum_{(I,J)\in\Gamma(k)}f^{(k)}_{I,J}x^Iy^J$, $1\leqslant k\leqslant M$,

$$\sum_{(I,J)\in\Gamma(k)}f^{(k)}_{I,J}u_{i+I,j+J}=0,\qquad (i,j)\in Z^2. \tag{2}$$

From Definition 3 we have immediately the following lemma.

Lemma 1: If $\mathfrak{A}$ and $\mathfrak{B}$ are ideals in $\tilde{R}$ such that $\mathfrak{A}\supset\mathfrak{B}$, then $G(\mathfrak{A})\subset G(\mathfrak{B})$.

Provided that there exist integers r,s such that $\mathfrak{A}\supset(x^r-1,y^s-1)$,[2] the partial periods of $\mathfrak{A}$ [14] are defined as follows:

$$\mathrm{per}_x(\mathfrak{A})=\min\{r>0|x^r-1\in\mathfrak{A}\}$$
$$\cdot\mathrm{per}_y(\mathfrak{A})=\min\{s>0|y^s-1\in\mathfrak{A}\}.$$

Lemma 2: For an arbitrary array u in $G(\mathfrak{A})$, $\mathrm{per}_x(u)|\mathrm{per}_x(\mathfrak{A})$, $\mathrm{per}_y(u)|\mathrm{per}_y(\mathfrak{A})$, provided that $\mathfrak{A}$ has finite partial periods.

It is easy to see the validity of the following relation.

Lemma 3: For the sum $\mathfrak{A}+\mathfrak{B}$ of ideals $\mathfrak{A},\mathfrak{B}$ in $\tilde{R}$, $G(\mathfrak{A})\cap G(\mathfrak{B})=G(\mathfrak{A}+\mathfrak{B})$.

III. Linear Recurring Arrays and Doubly Periodic Arrays

In this section we give the complete characterization of DP arrays as LR arrays. The following theorem ensures that we have only to examine polynomial ideals in order to investigate the properties of DP arrays.

Theorem 1 (Fundamental Theorem on Doubly Periodic and Linear Recurring Arrays): Let U be a finite nonempty set of arrays. U coincides with the set $G(\mathfrak{A})$ of LR arrays for a certain zero-dimensional ideal $\mathfrak{A}$ in $\tilde{R}$, i.e., an ideal with the finite algebraic set $V(\mathfrak{A})$,[3] if and only if U is an $\tilde{R}$-module of DP arrays.

We begin with treating the $\tilde{R}$-module $G(\mathfrak{A})$ of LR arrays.

Lemma 4: Provided that $\mathfrak{A}$ is a zero-dimensional ideal in $\tilde{R}$, $G(\mathfrak{A})$ is a finite subset of the set W of all DP arrays.

Proof: We may choose polynomials $f^{(1)},\cdots,f^{(M)}$ forming a basis of $\mathfrak{A}$. Eliminating y from the system of algebraic equations

$$f^{(1)}(x,y)=0,\cdots,f^{(M)}(x,y)=0,$$

we obtain the following system [17, pp. 1–5]: $D^{(1)}(x)=0,\cdots,D^{(N)}(x)=0$. Polynomials $D^{(1)}(x),\cdots,D^{(N)}(x)$ are elements of $\mathfrak{A}$. Since $\mathfrak{A}$ is zero-dimensional, for some k $D(x)=D^{(k)}(x)$ does not vanish identically. $D(x)$ can be assumed to have a nonzero constant term. Then there is a positive integer r such that $D(x)|x^r-1$. Therefore $\mathfrak{A}\ni x^r-1$. In the same way $\mathfrak{A}\ni y^s-1$ for a certain postive integer s. Thus any array u in $G(\mathfrak{A})$ is doubly periodic with partial periods such that $\mathrm{per}_x(u)|r$, $\mathrm{per}_y(u)|s$. Q.E.D.

For a one-dimensional ideal $\mathfrak{A}$ in $\tilde{R}$, there exist arrays in $G(\mathfrak{A})$ which have a finite P. vector only along a single direction. From now on we exclude such cases, dealing only with zero-dimensional ideals. A finite $\tilde{R}$-module $G(\mathfrak{A})$ is a finite-dimensional F-module.

Definition 4: Let $\dim G(\mathfrak{A})$ denote the dimension of the F-module $G(\mathfrak{A})$.

The linear recurrences (2) can be regarded as a system of linear equations with respect to the rs components $u_{i,j}$, $0\leqslant i\leqslant r-1$, $0\leqslant j\leqslant s-1$,

$$\sum_{i=0}^{r-1}\sum_{j=0}^{s-1}f^{(k,I,J)}_{i,j}u_{i,j}=0,\qquad 0\leqslant I\leqslant r-1,\ 0\leqslant J\leqslant s-1,\ 1\leqslant k\leqslant M \tag{3}$$

where

$$f^{(k,I,J)}_{i,j}=\begin{cases}f^{(k)}_{i-I,j-J}, & \text{if } (i-I,j-J)\in\Gamma(k)\\ 0, & \text{otherwise}\end{cases}$$

and the subscripts i,j are to be interpreted respectively mod r and mod s. The polynomials $f^{(k,I,J)}(x,y)\triangleq\sum_{i=0}^{r-1}\sum_{j=0}^{s-1}f^{(k,I,J)}_{i,j}x^iy^j$ can be regarded as elements of the residue class ring $\mathfrak{A}/(x^r-1,y^s-1)$. Clearly, $\dim G(\mathfrak{A})$ is equal to the dimension of the solution space of the linear system (3). Since the rank of (3) is equal to the dimension $\dim_F(\mathfrak{A}/(x^r-1,y^s-1))$ of the residue class ring $\mathfrak{A}/(x^r-1,y^s-1)$ as an F-module, we have from Lemma B1 (Appendix B),

$$\begin{aligned}\dim G(\mathfrak{A})&=rs-\dim_F(\mathfrak{A}/(x^r-1,y^s-1))\\&=\dim_F(\tilde{R}/(x^r-1,y^s-1))\\&\qquad-\dim_F(\mathfrak{A}/(x^r-1,y^s-1))\\&=\dim_F(\tilde{R}/\mathfrak{A}).\end{aligned} \tag{4}$$

[1] $a|b$ means that there exists an integer c such that $ac=b$.

[2] Equivalently, $\mathfrak{A}$ is a zero-dimensional ideal. (See Lemma 4.)

[3] The algebraic set $V(\mathfrak{A})$ is defined to be the totality of all zeros common to the polynomials contained in $\mathfrak{A}$. The smallest possible number of parameters which are needed in the parametric representation of an algebraic set $V(\mathfrak{A})$ is called the dimension of $V(\mathfrak{A})$ or the *dimension* of $\mathfrak{A}$ [17, pp. 46–71].

Authorized licensed use limited to: Lakehead University. Downloaded on September 25,2026 at 17:14:10 UTC from IEEE Xplore. Restrictions apply.

It should be noted that (4) is valid irrespective of the choice of integers r,s as long as $\mathfrak{A} \supset (x^r-1, y^s-1)$. Thus the F-module $G(\mathfrak{A})$ is (F-) isomorphic to the F-module $\tilde{R}/\mathfrak{A}$.

The $\tilde{R}$-module $G(\mathfrak{A})$ is also an $\tilde{R}/\mathfrak{A}$-module. More generally we have the following lemma.

Lemma 5: Provided that the ideal $\mathfrak{B}$ is a subset of the ideal $\mathfrak{A}$, the difference module $G(\mathfrak{B})-G(\mathfrak{A})$ is a $\mathfrak{A}/\mathfrak{B}$-module with its dimension as an F-module equal to that of $\mathfrak{A}/\mathfrak{B}$.

If $\mathfrak{A} \supsetneq \mathfrak{B}$, then $\dim\ G(\mathfrak{A}) < \dim\ G(\mathfrak{B})$, since $\tilde{R}/\mathfrak{A} \subsetneq R/\mathfrak{B}$. From this remark we have the converse to Lemma 1.

Lemma 6: If $G(\mathfrak{A}) \subset G(\mathfrak{B})$, then $\mathfrak{A} \supset \mathfrak{B}$.

Proof: Suppose $G(\mathfrak{A}) \subset G(\mathfrak{B})$ and $\mathfrak{A} \not\supset \mathfrak{B}$. Then $G(\mathfrak{A}) = G(\mathfrak{A}) \cap G(\mathfrak{B}) = G(\mathfrak{A}+\mathfrak{B})$. On the other hand, $\mathfrak{A}+\mathfrak{B} \supsetneq \mathfrak{A}$, from which it follows that $G(\mathfrak{A}) \supsetneq G(\mathfrak{A}+\mathfrak{B})$. This is a contradiction. Q.E.D.

Let $K = \dim G(\mathfrak{A})$. Then there is a unique array in $G(\mathfrak{A})$ with an arbitrary set of values $u_{i,j}$ in F for (i,j) in a certain subset Γ of Z^2 such that $|\Gamma| = K$. Such a subset Γ of Z^2 is said to be an *independent point set* of the ideal $\mathfrak{A}$ or of $G(\mathfrak{A})$ and is denoted by $\Delta(\mathfrak{A})$. Γ may be obtained from a canonical form of the homogeneous linear system (3). From the preceding considerations we have the following theorem.

Theorem 2: A subset Γ of Z^2 is an independent point set of a (zero-dimensional) ideal $\mathfrak{A}$ if and only if

i) $|\Gamma| = \dim G(\mathfrak{A})$;
ii) $\mathfrak{A}$ does not contain a nonzero polynomial of the form $f(x,y) = \sum_{(i,j)\in\Gamma} f_{i,j} x^i y^j$.

For $(k,l) \notin \Delta(\mathfrak{A})$, there is an element $g^{(k,l)}(x,y)$ of $\mathfrak{A}$ such that $g^{(k,l)}(x,y) = x^k y^l + \sum_{(i,j)\in\Delta(\mathfrak{A})} g_{i,j}^{(k,l)} x^i y^j$. Thus for a given set of values $u_{i,j} \in F((i,j) \in \Delta(\mathfrak{A}))$, we can determine the value $u_{k,l}$ for any point $(k,l) \notin \Delta(\mathfrak{A})$ from

$$u_{k,l} = -\sum_{(i,j)\in\Delta(\mathfrak{A})} g_{i,j}^{(k,l)} u_{i,j}.$$

Note that an independent point set as defined by Ikai *et al.* [16] is a special case of Theorem 2.

Example 2: Let $q=2$. The system of LR relations:

$$\begin{cases} w_{i+4,j} + w_{i+2,j} + w_{i,j} = 0 \\ w_{i+2,j+1} + w_{i+1,j+1} + w_{i,j+1} + w_{i+3,j} + w_{i+2,j} + w_{i+1,j} = 0 \\ w_{i,j+3} + w_{i+1,j+2} + w_{i,j+2} + w_{i+1,j+1} + w_{i,j+1} + w_{i+1,j} = 0 \end{cases} \tag{5}$$

corresponds to the ideal $\mathfrak{A} = (x^4+x^2+1,\ (x^2+x+1)(y+x),\ y^3+(x+1)y^2+(x+1)y+x)$. For a root α of x^2+x+1 in the extension field GF (2^2), the algebraic set is $V(\mathfrak{A}) = \{(\alpha,\alpha),(\alpha^2,\alpha^2),(\alpha,\alpha^2),(\alpha^2,\alpha)\}$. A LR array w satisfying (5) is shown in Fig. 3, where $\mathrm{per}_x\ (w) = \mathrm{per}_y\ (w) = 6$, per $(w) = 36$. $\Delta(\mathfrak{A}) = \{(0,0), (0,1), (0,2), (1,0), (1,1), (1,2), (2,1), (3,1)\}$ is an independent point set and $\dim G(\mathfrak{A}) = 8$. From the values in $\Delta(\mathfrak{A})$, the remaining values are determined

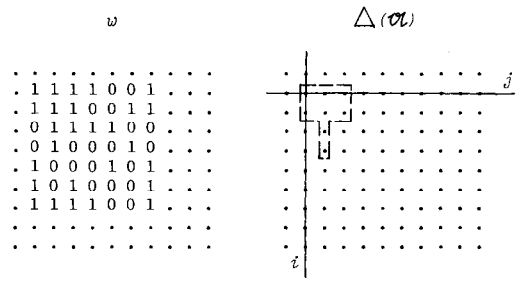


Fig. 3. Example of LR array over GF (2) and independent point set.

recursively by

$$\begin{aligned} &(x^2+y(x^3+1)+x+1)w \\ &\quad = (x^4y+(x^2+1)y)w \\ &\quad = (x^2y^2+y^2(x+1)+y(x^3+x^2+x))w \\ &\quad = (y^3+(x+1)y^2+(x+1)y+x)w = (0), \end{aligned}$$

which can be derived from (5).

For any finite set of DP arrays, there exists a corresponding ideal.

Theorem 3: If U is a finite nonempty subset of W_∞ then there exists a unique (zero-dimensional) ideal $\mathfrak{A}$ in $\tilde{R}$ such that an ideal $\mathfrak{B}$ is contained in $\mathfrak{A}$ if and only if U is contained in $G(\mathfrak{B})$. The ideal $\mathfrak{A}$ coincides with the set $\mathfrak{A}(U) \triangleq \{f(x,y) \in \tilde{R} \mid f(x,y)u = (0),\ u \in U\}$.

Definition 5: The ideal $\mathfrak{A}(U)$ is said to be the *maximum ideal* associated with U or the *characteristic ideal* of U. If U contains a single array u, $\mathfrak{A}(\{u\})$ is also said to be the characteristic ideal $\mathfrak{A}(u)$ of u.[4]

$\mathfrak{A}(u)$ may be compared with the ideal generated by the minimum polynomial of a periodic sequence. As for the partial periods of $\mathfrak{A}(u)$, we have, in view of Lemma 1 and the maximality of $\mathfrak{A}(u)$ for u, the following lemma and its corollaries.

Lemma 7: Let $\mathfrak{A} = \mathfrak{A}(u)$; then $\mathrm{per}_x\ (u) = \mathrm{per}_x\ (\mathfrak{A})$, and similarly for y.

Corollary 1: $\mathrm{per}_x\ (\mathfrak{A}) = \max\ \{\mathrm{per}_x\ (u) \mid u \in G(\mathfrak{A})\}$, and similarly for y.

Corollary 2: Provided that $\mathfrak{P}$ is a (zero-dimensional) prime ideal, for any nonzero array u in $G(\mathfrak{P})$, $\mathrm{per}_x\ (u) = \mathrm{per}_x\ (\mathfrak{P})$, and similarly for y.

Both $\mathrm{per}_x\ (\mathfrak{P})$ and $\mathrm{per}_y\ (\mathfrak{P})$ are relatively prime to the characteristic p of the field F. There is the following relation between a primary ideal $\mathfrak{Q}$ and the prime ideal $\mathfrak{P}$ belonging to $\mathfrak{Q}$ [17, pp. 26–30]: for some integer l, $\mathfrak{P} \supset \mathfrak{Q} \supset \mathfrak{P}^l$. It follows that $\mathrm{per}_x\ (\mathfrak{Q}) = \mathrm{per}_x\ (\mathfrak{P})p^m$, $\mathrm{per}_y\ (\mathfrak{Q}) = \mathrm{per}_y\ (\mathfrak{P})p^n$, where m and n are integers such that $0 \leqslant p^m, p^n \leqslant l$.

The sum $G(\mathfrak{A}) + G(\mathfrak{B})$ of $\tilde{R}$-modules $G(\mathfrak{A})$ and $G(\mathfrak{B})$ is an $\tilde{R}$-submodule of $G(\mathfrak{A} \cap \mathfrak{B})$ and of $G(\mathfrak{A} \cdot \mathfrak{B})$. The following result corresponds to that of Ikai *et al.* [3, th. 8].

Theorem 4: $G(\mathfrak{A}) + G(\mathfrak{B}) = G(\mathfrak{A} \cap \mathfrak{B})$.

[4] A method of determining the characteristic ideal $\mathfrak{A}(u)$ of a given DP array u is described in Appendic C.

Authorized licensed use limited to: Lakehead University. Downloaded on September 25,2026 at 17:14:10 UTC from IEEE Xplore. Restrictions apply.

Proof: From Lemma B2 (Appendix B), we have $\dim_F (\mathfrak{A}+\mathfrak{B}/\mathfrak{B}) = \dim_F (\mathfrak{A}/\mathfrak{A}\cap\mathfrak{B})$, from which it follows, in view of Lemma 5, that $\dim G(\mathfrak{B}) - \dim G(\mathfrak{A}+\mathfrak{B}) = \dim G(\mathfrak{A}\cap\mathfrak{B}) - \dim G(\mathfrak{A})$. Accordingly

$$\begin{aligned}&\dim (G(\mathfrak{A}) + G(\mathfrak{B}))\\ &\quad = \dim G(\mathfrak{A}) + \dim G(\mathfrak{B}) - \dim (G(\mathfrak{A})\cap G(\mathfrak{B}))\\ &\quad = \dim G(\mathfrak{A}) + \dim G(\mathfrak{B}) - \dim G(\mathfrak{A}+\mathfrak{B})\\ &\quad = \dim G(\mathfrak{A}\cap\mathfrak{B}),\end{aligned}$$

from which it follows that $G(\mathfrak{A}) + G(\mathfrak{B}) = G(\mathfrak{A}\cap\mathfrak{B})$. Q.E.D.

Provided that two ideals $\mathfrak{A},\mathfrak{B}$ are relatively prime, the $\tilde{R}$-module $G(\mathfrak{A}\cap\mathfrak{B}) = G(\mathfrak{A}\cdot\mathfrak{B})$ is a direct sum of $\tilde{R}$-modules $G(\mathfrak{A})$ and $G(\mathfrak{B})$, because $G(\mathfrak{A})\cap G(\mathfrak{B}) = G(\mathfrak{A}+\mathfrak{B}) = G(\tilde{R}) = \{(0)\}$. A (zero-dimensional) ideal $\mathfrak{A}$ can be represented as the intersection of greatest primary components (Appendix A): $\mathfrak{A} = \cap_i \mathfrak{A}_i$. Then the algebraic set is $V(\mathfrak{A}) = \cup_i V(\mathfrak{A}_i) = \cup_i V(\mathfrak{P}_i)$, where $\mathfrak{P}_i$ are distinct zero-dimensional prime ideals, each $\mathfrak{P}_i$ belonging to $\mathfrak{A}_i$. Because two distinct zero-dimensional prime ideals $\mathfrak{P}_1,\mathfrak{P}_2$ in $\tilde{R}$ are relatively prime, the primary ideals $\mathfrak{A}_1,\mathfrak{A}_2$, respectively, belonging to $\mathfrak{P}_1,\mathfrak{P}_2$, are also. Therefore, the above decomposition can be rewritten as $\mathfrak{A} = \cap_i \mathfrak{A}_i = \Pi_i \mathfrak{A}_i$.

Corollary 3: $G(\mathfrak{A})$ can be expressed as the direct sum

$$G(\mathfrak{A}) = G(\mathfrak{A}_1) \dot{+} \cdots \dot{+} G(\mathfrak{A}_M).$$

Example 3: The ideal $\mathfrak{A}$ in Example 2 can be decomposed into greatest primary components as follows: $\mathfrak{A} = \mathfrak{A}_1\cdot\mathfrak{A}_2$, where $\mathfrak{A}_1 = (x^2+x+1, y+x+1)$ is the prime ideal with $V(\mathfrak{A}_1) = \{(\alpha,\alpha^2),(\alpha^2,\alpha)\}$ and $\mathfrak{A}_2 = (x^4+x^2+1, (x^2+x+1)y+x^3+x^2+x, y^2+x^2)$ is a primary ideal belonging to the prime ideal $\mathfrak{P}_2 = (x^2+x+1, y+x)$ with $V(\mathfrak{A}_2) = V(\mathfrak{P}_2) = \{(\alpha,\alpha),(\alpha^2,\alpha^2)\}$. The array w (Fig. 3) can be represented uniquely by the sum of u (Fig. 1) in $G(\mathfrak{A}_1)$ and v (Fig. 2) in $G(\mathfrak{A}_2)$. $\mathfrak{A}_2$ may be decomposed as $\mathfrak{A}_2 = \mathfrak{Q}\cap\mathfrak{Q}'$, where $\mathfrak{Q} = (x^4+x^2+1, y+x), \mathfrak{Q}' = (x^2+x+1, y^2+x^2)$ are primary ideals belonging to $\mathfrak{P}_2$. But $G(\mathfrak{Q}) + G(\mathfrak{Q}')$ is not a direct sum.

Corollary 4: Provided that $\mathfrak{A}$ and $\mathfrak{B}$ are relatively prime ideals, $\mathrm{per}_x (u+v) = \mathrm{lcm}\,(\mathrm{per}_x (u), \mathrm{per}_x (v))$, for $u\in G(\mathfrak{A})$, $v\in G(\mathfrak{B})$, and similarly for y.

We now treat a finite $\tilde{R}$-module of DP arrays.

Lemma 8: Let U be any finite $\tilde{R}$-module which is a subset of W. If u is an array in U, then $G(\mathfrak{A})\subset U$, where $\mathfrak{A} = \mathfrak{A}(u)$ is the characteristic ideal of u.

Proof: Note that u and its translates ${}_l u$ are contained in U. Let $\Delta(\mathfrak{A})$ denote an independent point set of $\mathfrak{A}$. Suppose the translates ${}_l u$ for $l\in\Delta(\mathfrak{A})$ are linearly dependent. Then there must be a nonzero polynomial $f(x,y) = \Sigma_{(I,J)\in\Delta(\mathfrak{A})} f_{I,J} x^I y^J$ such that $f(x,y)u = (0)$, which implies that $u\in G(\mathfrak{A}+(f(x,y)))$. By the definition of $\mathfrak{A}$, $f(x,y)$ must be an element of $\mathfrak{A}$, which contradicts Theorem 2. Consequently the F-module $G(\mathfrak{A})$ is spanned by the translates ${}_l u$ for $l\in\Delta(\mathfrak{A})$, the cardinality of which is equal to $\dim G(\mathfrak{A})$. Therefore, $G(\mathfrak{A})$ is a submodule of U. Q.E.D.

In view of Theorem 4 and Lemma 8, we can complete the proof of Theorem 1 (see the beginning of this section).

Proof of Theorem 1: Necessity is already trivial. Let $u_1,\cdots,u_M$ be the elements of U and $\mathfrak{A}_k$ be the characteristic ideal of each u_k, $1\leqslant k\leqslant M$. Then $G(\mathfrak{A}_k)\subset U$ from Lemma 8 and $G(\mathfrak{A}_1) + \cdots + G(\mathfrak{A}_M)\subset U$. But $U\subset G(\mathfrak{A}_1) + \cdots + G(\mathfrak{A}_M)$, since $u_k\in G(\mathfrak{A}_k)$, $1\leqslant k\leqslant M$. Therefore, $U = G(\mathfrak{A}_1) + \cdots + G(\mathfrak{A}_M) = G(\mathfrak{A})$ from Theorem 4, where $\mathfrak{A} = \mathfrak{A}_1\cap\cdots\cap\mathfrak{A}_M$. Q.E.D.

From Theorem 1 we also obtain the one-to-one correspondence between $\tilde{R}$-modules of DP arrays and two-dimensional cyclic codes, which are special cases of Abelian codes.

IV. Equivalent Doubly Periodic Arrays

Given an integral unimodular matrix $\boldsymbol{S}$, the array u' such that $u'(\boldsymbol{i}) = u(\boldsymbol{iS})$, $\boldsymbol{i}\in Z^2$, is said to be the array $\phi_{\boldsymbol{S}}(u)$ *rearranged from* u by means of the unimodular transformation $\boldsymbol{S}$.

Lemma 9: For $\boldsymbol{kS} = \boldsymbol{m}$, $\phi_{\boldsymbol{S}}({}_{\boldsymbol{m}}u) = {}_{\boldsymbol{k}}\phi_{\boldsymbol{S}}(u)$.

Definition 6: If two arrays u,v are transformed into each other by rearrangement and/or translation, they are said to be *equivalent* to each other.

If an array u may be rearranged to give another array u' by means of $\boldsymbol{T}$, a P. vector $\boldsymbol{l}$ of u is transformed to a P. vector $\boldsymbol{l}'$ of u' by $\boldsymbol{l}' = \boldsymbol{lT}^{-1}$. Thus in view of (1), the following relation holds between the FP parallelogram $\boldsymbol{l}_1\times\boldsymbol{l}_2$ of u and the FP parallelogram $\boldsymbol{m}_1\times\boldsymbol{m}_2$ of $u' = \phi_{\boldsymbol{T}}(u)$:

$$\begin{Vmatrix}\boldsymbol{m}_1\\ \boldsymbol{m}_2\end{Vmatrix} = \boldsymbol{S}\begin{Vmatrix}\boldsymbol{l}_1\\ \boldsymbol{l}_2\end{Vmatrix}\boldsymbol{T}^{-1} \tag{6}$$

where $\boldsymbol{S}$ and $\boldsymbol{T}$ are unimodular matrices.

Lemma 10: A DP array has the same period as any array equivalent to it. The cycle containing u is transformed to the cycle containing an array equivalent to u.

In view of (6), we have the following theorem.

Theorem 5: An arbitrary DP array with a FP parallelogram $\boldsymbol{l}_1\times\boldsymbol{l}_2$ has an equivalent array of which a FP parallelogram $\boldsymbol{l}_1'\times\boldsymbol{l}_2'$ is a rectangle with $\boldsymbol{l}_1' = (r,0)$, $\boldsymbol{l}_2' = (0,s)$, where r and s are elementary divisors of the integral matrix

$$\begin{Vmatrix}\boldsymbol{l}_1\\ \boldsymbol{l}_2\end{Vmatrix}.$$

Example 4: Let v' be the array rearranged from v (Fig. 2) by means of

$$\boldsymbol{T} = \begin{Vmatrix}2 & 1\\ -1 & 0\end{Vmatrix}.$$

Then v' has a FP parallelogram $\boldsymbol{l}_1'\times\boldsymbol{l}_2'$, where

$$\begin{Vmatrix}\boldsymbol{l}_1'\\ \boldsymbol{l}_2'\end{Vmatrix} = \begin{Vmatrix}2 & 0\\ 0 & 6\end{Vmatrix} = \boldsymbol{S}\begin{Vmatrix}4 & 2\\ 2 & 4\end{Vmatrix}\boldsymbol{T}^{-1}, \qquad \text{for } \boldsymbol{S} = \begin{Vmatrix}1 & 0\\ -2 & 1\end{Vmatrix},$$

Authorized licensed use limited to: Lakehead University. Downloaded on September 25,2026 at 17:14:10 UTC from IEEE Xplore. Restrictions apply.

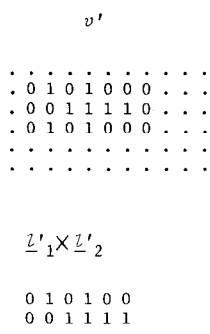


Fig. 4. Example of rearranged array and its FP parallelogram.

and per (v)=per (v')=12, per$_x$ (v')=2, per$_y$ (v')=6. (See Fig. 4)

Clearly the number of times that each element of F appears in any FP parallelogram $\boldsymbol{l}_1 \times \boldsymbol{l}_2$ of a DP array is left invariant under rearrangement by means of the elementary unimodular transformations

$$\begin{Vmatrix} -1 & 0 \\ 0 & 1 \end{Vmatrix} \quad \begin{Vmatrix} 1 & 0 \\ 0 & -1 \end{Vmatrix} \quad \begin{Vmatrix} 1 & 1 \\ 0 & 1 \end{Vmatrix}$$

which generate the unimodular transformation group. This yields the following lemma. The *weight* of $\boldsymbol{l}_1 \times \boldsymbol{l}_2$ is defined to be the number of nonzero values on the points contained in $\boldsymbol{l}_1 \times \boldsymbol{l}_2$.

Lemma 11: A FP parallelogram of a DP array has the same weight as that of any equivalent array.

Corresponding to $\boldsymbol{l}_1 \times \boldsymbol{l}_2$ with

$$\begin{Vmatrix} \boldsymbol{l}_1 \\ \boldsymbol{l}_2 \end{Vmatrix} = \begin{Vmatrix} r & 0 \\ r'' & s'' \end{Vmatrix} \text{ or } \begin{Vmatrix} r' & s' \\ 0 & s \end{Vmatrix},$$

the set of lattice points $\Gamma=\{(i,j)|0\leqslant i<r,\ 0\leqslant j<s''\}$ or $\{(i,j)|0\leqslant i<r',\ 0\leqslant j<s\}$ is said to be a *unit rectangle* of u. A DP array has the same distribution of elements of F in a unit rectangle as in a FP parallelogram. This implies that equivalent DP arrays corresponds to equivalent codewords. One of the aims of the following is to study the equivalence between Abelian group codes.

From the linearity of rearrangement ϕ_S and since $\phi_S(G(\mathfrak{A}))$ is closed under translations, $\phi_S(G(\mathfrak{A}))$ is an $\tilde{R}$-module of DP arrays. From Theorem 1, $\phi_S(G(\mathfrak{A}))=G(\mathfrak{A}^S)$ for a certain ideal $\mathfrak{A}^S$ in $\tilde{R}$. Thus ϕ_S induces a ring-automorphism of $\tilde{R}$. Any automorphism of $\tilde{R}$ is of the form: $\psi_S(f(x,y))=f(x^a y^b, x^c y^d)$, where

$$\boldsymbol{S}=\begin{Vmatrix} a & b \\ c & d \end{Vmatrix}$$

is a unimodular matrix.

Theorem 6: $\phi_S(G(\mathfrak{A}))=G(\psi_S^{-1}(\mathfrak{A}))$.

Corollary 5: dim $G(\mathfrak{A})$=dim $G(\psi_S^{-1}(\mathfrak{A}))$, i.e., dim $G(\mathfrak{A})$ =dim $\phi_S(G(\mathfrak{A}))$.

Example 5: Let $q=2$. Consider $\mathfrak{A}=(x^4+x^2+1, (x^2+x+1)(y+x), y^2+x^2)$ $(=\mathfrak{A}_2$ in Example 3). For

$$\boldsymbol{T}=\begin{Vmatrix} 2 & 1 \\ -1 & 0 \end{Vmatrix},$$

$$\mathfrak{B} \triangleq \psi_T^{-1}(\mathfrak{A})=(y^{-4}+y^{-2}+1, (y^{-2}+y^{-1}+1)(xy^2+y^{-1}), x^2y^4+y^{-2})$$
$$=(y^4+y^2+1, (y^2+y+1)(x+1), x^2+1),$$

where dim $G(\mathfrak{A})$=dim $G(\mathfrak{B})=6$.

So far we have considered only linear invertible mappings ϕ_S corresponding to unimodular matrices $\boldsymbol{S}$. We now introduce a generalization of rearrangement and of decimation as defined by Nomura *et al.* [2].

Definition 7: For a nonsingular integral matrix

$$\boldsymbol{A}=\begin{Vmatrix} a & b \\ c & d \end{Vmatrix},$$

the array $v=\phi_A(u)$ such that $v(\boldsymbol{i})=u(\boldsymbol{iA})$, $\boldsymbol{i}\in Z^2$, is said to be the *A-decimated array* of u.

Lemma 9 is also valid for any decimation ϕ_A. By analogy with the ring-automorphism ψ_S, we introduce the endomorphism ψ_A for a nonsingular integral matrix $\boldsymbol{A}$.

Definition 8: For

$$\boldsymbol{A}=\begin{Vmatrix} a & b \\ c & d \end{Vmatrix},$$

det $(\boldsymbol{A})\neq 0$, and for $f(x,y)\in\tilde{R}$, $\psi_A(f(x,y))=f'(x,y)\triangleq f(x^a y^b, x^c y^d)$.

ψ_A is a unitary injective endomorphism of $\tilde{R}$. If $\mathfrak{A}$ is an ideal in $\tilde{R}$, so is the inverse image $\psi_A^{-1}(\mathfrak{A})$. Thus we have an extension of Theorem 6.

Theorem 7: $\phi_A(G(\mathfrak{A}))=G(\psi_A^{-1}(\mathfrak{A}))$.

Proof: Let $v=\phi_A(u)$ for an arbitrary array u in $G(\mathfrak{A})$, and consider the operation $f(x,y)=\sum_{(k,l)\in\Gamma(f)} f_{k,l}x^k y^l \in \psi_A^{-1}(\mathfrak{A})$ on the array v. We have the identity $\sum_{\boldsymbol{k}\in\Gamma(f)} f_{\boldsymbol{k}} v(\boldsymbol{i}+\boldsymbol{k})=\sum_{\boldsymbol{k}\in\Gamma(f)} f_{\boldsymbol{k}} u((\boldsymbol{i}+\boldsymbol{k})\boldsymbol{A})$, which is the $\boldsymbol{i}'$-component of the array $f'(x,y)u$ and which must vanish identically, where $\boldsymbol{i}'=\boldsymbol{iA}$ and $f'(x,y)=\psi_A(f(x,y))\in\mathfrak{A}$. Conversely, define u by $u(\boldsymbol{iA}+\boldsymbol{m})=v(\boldsymbol{i})$ for an array $v\in G(\psi_A^{-1}(\mathfrak{A}))$, where $\boldsymbol{m}$ runs through each coset of the additive group Z^2 modulo the subgroup $Z_A^2 \triangleq \{\boldsymbol{i}'|\boldsymbol{i}'=\boldsymbol{iA},\ \boldsymbol{i}\in Z^2\}$. Thus u is uniquely defined from v by a choice of the coset leaders $\boldsymbol{m}$. Then for $f(x,y)\in\psi_A^{-1}(\mathfrak{A})$, we have the identity $0=\sum_{\boldsymbol{k}\in\Gamma(f)} f_{\boldsymbol{k}} v(\boldsymbol{i}+\boldsymbol{k})=\sum_{\boldsymbol{k}\in\Gamma(f)} f_{\boldsymbol{k}} u((\boldsymbol{i}+\boldsymbol{k})\boldsymbol{A}+\boldsymbol{m})$, where the set of points $\{\boldsymbol{i}'|\boldsymbol{i}'=\boldsymbol{iA}+\boldsymbol{m}\}$ coincides with Z^2. Therefore, we have $f'(x,y)u=(0)$, where $f'(x,y)=\psi_A(f(x,y))\in\mathfrak{A}$. Thus $u\in G(\mathfrak{A})$, and $v=\phi_A({}_m u)\in\phi_A(G(\mathfrak{A}))$. Q.E.D.

Example 6: Consider $v\in G(\mathfrak{A})$, $\mathfrak{A}=(x^4+x^2+1, (x^2+x+1)\cdot(y+x), y^2+x^2)$. (See Fig. 2.) For

$$\boldsymbol{A}=\begin{Vmatrix} 2 & 1 \\ -1 & 1 \end{Vmatrix}=\begin{Vmatrix} 1 & 0 \\ 1 & 1 \end{Vmatrix}\cdot\begin{Vmatrix} 1 & 0 \\ 0 & 3 \end{Vmatrix}\cdot\begin{Vmatrix} 2 & 1 \\ -1 & 0 \end{Vmatrix},$$

$v'=\phi_A(v)$ and $v''=\phi_A({}_{1,0}v)\in G(\mathfrak{B})$, $\mathfrak{B}\triangleq\psi_A^{-1}(\mathfrak{A})=(x^2+1, y^2+1, xy+y+x+1)$, are shown in Fig. 5, where dim $G(\mathfrak{B})=3$.

Authorized licensed use limited to: Lakehead University. Downloaded on September 25,2026 at 17:14:10 UTC from IEEE Xplore. Restrictions apply.

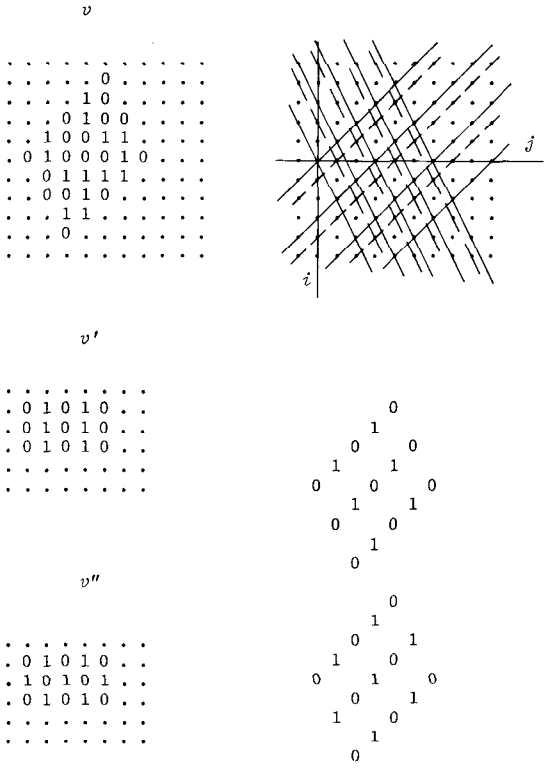


Fig. 5. Examples of decimated arrays.

Clearly, we have $\phi_A(G(\mathfrak{A}) + G(\mathfrak{B})) = \phi_A(G(\mathfrak{A})) + \phi_A(G(\mathfrak{B}))$. In the above proof, we have introduced an operation which is dual to decimation ϕ_A. Generalizing it, we define two-dimensional interleaving.

Definition 9: The arrays in the set $\phi_A^*(G(\mathfrak{A})) \triangleq \{u | \phi_A({}_i u) \in G(\mathfrak{A}),\ i \in Z^2\}$ are called *A-interleaved arrays* of $G(\mathfrak{A})$.

Provided S is a unimodular matrix, $\phi_A^* = \phi_S^{-1} = \phi_{S^{-1}}$. In general, as operations on sets of arrays

$$\phi_A \cdot \phi_A^* = 1, \tag{7}$$

but $\phi_A^* \cdot \phi_A \neq 1$. The following theorem is proved in Appendix D.

Theorem 8: $\phi_A^*(G(\mathfrak{A})) = G(\mathfrak{A}_A)$, where $\mathfrak{A}_A$ is the minimum ideal that satisfies $\psi_A^{-1}(\mathfrak{A}_A) = \mathfrak{A}$ and contains $\psi_A(\mathfrak{A})$. Furthermore, dim $G(\mathfrak{A}_A)$ is equal to dim $G(\mathfrak{A})$ multiplied by the absolute value of det (A).

Example 7: Consider $\mathfrak{A} = (x^4 + x^2 + 1,\ (x^2 + x + 1)(y + x),\ y^2 + x^2)$ in Example 6. For

$$B = \begin{Vmatrix} 1 & 1 \\ -1 & 1 \end{Vmatrix},$$

$\mathfrak{A}_B = (x^4 + 1,\ (x^2 + 1)y^2 + (x + x^3)y + x^2 + 1,\ y^4 + y^2 + (x + x^3)y + x^2)$, where dim $G(\mathfrak{A}_B) = 12 = 6 \cdot 2$, det $(B) = 2$ and dim $G(\mathfrak{A}) = 6$. An interleaved array $v^* \in G(\mathfrak{A}_B)$ is shown in Fig. 6. A translate of $\phi_B(v^*)$ yields $v \in G(\mathfrak{A})$ in Figs. 2 and 5. For $\mathfrak{B}$ in Example 6, $\mathfrak{A}$ is not the minimum ideal containing $\psi_A(\mathfrak{B})$, but $\mathfrak{B}_A = (x^6 + 1,\ (x^3 + 1)y + x^4 + x,\ y^2 + x^2)$ is. dim $G(\mathfrak{B}_A) = 9 = 3 \cdot 3$, and the interleaved arrays of $G(\mathfrak{B})$ are $G(\mathfrak{B}_A)$, but not $G(\mathfrak{A})$.

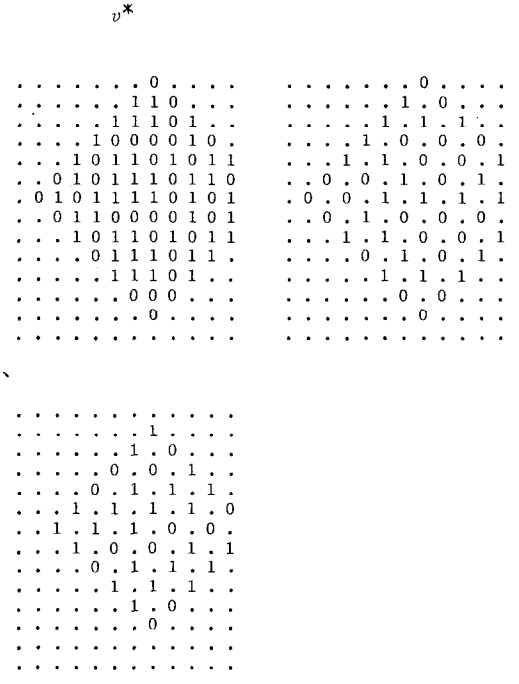


Fig. 6. Example of interleaved array.

V. Irreducible Linear Recurring Arrays and M-Arrays

Since a zero-dimensional prime ideal $\mathfrak{P}$ is maximal in $\tilde{R}$, $G(\mathfrak{P})$ is a minimal $\tilde{R}$-module of DP arrays (except for the trivial module $\{(0)\}$). An arbitrary nonzero array in a minimal $\tilde{R}$-module $G(\mathfrak{P})$ is said to be *irreducible*.

The algebraic set of $\mathfrak{P}$ consists of algebraically conjugate zeros over $F = \mathrm{GF}(q)$: $V(\mathfrak{P}) = \{(\alpha, \beta), (\alpha^q, \beta^q), \cdots, (\alpha^{q^{K-1}}, \beta^{q^{K-1}})\}$, where α, β are nonzero elements of an extension field GF (q^L) of F, and $K | L$. Thus α, β can be expressed as powers of a primitive element θ of GF (q^L): $\alpha = \theta^\kappa$, $\beta = \theta^\lambda$. Then $K = |V(\mathfrak{P})| = \min\ \{K > 0 |\ \mu q^K \equiv \mu \bmod N\}$, where $\mu \triangleq \gcd(\kappa, \lambda)$ and $N \triangleq q^L - 1$. $\mathfrak{P}$ is generated by irreducible polynomials: $\mathfrak{P} = (f(x), h(x,y)) = (g(y), \tilde{h}(x,y))$. $K = \mathrm{lcm}\ (K_1, K_2)$, where $K_1 = \deg f$, the degree of $f(x)$, and $K_2 = \deg g$. The degree in y of $h(x,y)$ is $\deg_y h = K_2' = K/K_1$, and the degree in x of $\tilde{h}(x,y)$ is $\deg_x \tilde{h} = K_1' = K/K_2$. Both $\{(i,j) | 0 \leq i < K_1,\ 0 \leq j < K_2'\}$ and $\{(i,j) | 0 \leq i < K_1',\ 0 \leq j < K_2\}$ are independent point sets.

An arbitrary array u in $G(\mathfrak{P})$ is given as follows:

$$u_{i,j} = \sum_{0 \leq k \leq K-1} \gamma_k (\alpha^i \beta^j)^{q^k}, \qquad (i,j) \in Z^2 \tag{8}$$

where γ_k, $0 \leq k \leq K-1$, are elements of GF (q^L). More specifically, let $L = K$. Then a nonzero array is given as

$$u_{i,j} = \sum_{0 \leq k \leq K-1} \theta^{(M + \kappa i + \lambda j) q^k}, \qquad (i,j) \in Z^2 \tag{9}$$

where M is an integer such that $0 \leq M \leq q^K - 2$. Thus, as is easily shown, the partial periods of any nonzero array u in $G(\mathfrak{P})$ are obtained as $\mathrm{per}_x\ (u) = r \triangleq \min\ \{r > 0 | r\kappa \equiv 0 \bmod N\} = N/\gcd\ (N, \kappa)$, $\mathrm{per}_y\ (u) = s \triangleq N/\ \gcd\ (N, \lambda)$, where

Authorized licensed use limited to: Lakehead University. Downloaded on September 25,2026 at 17:14:10 UTC from IEEE Xplore. Restrictions apply.

$N \triangleq q^K - 1$. As for the period, we have the following result.

Theorem 9: $G(\mathfrak{P})$——$\{(0)\}$ consists of $(q^K-1)/t$ cycles of length $t \triangleq \text{lcm}\,(r,s) = \min\{t>0 \mid t\mu \equiv 0 \bmod q^K-1\} = (q^K-1)/\gcd(q^K-1,\mu)$.

A FP parallelogram $\boldsymbol{l}_1 \times \boldsymbol{l}_2$ of any nonzero array in $G(\mathfrak{P})$ is given by

$$\begin{Vmatrix} \boldsymbol{l}_1 \\ \boldsymbol{l}_2 \end{Vmatrix} = \begin{Vmatrix} r & 0 \\ r'' & s'' \end{Vmatrix} \text{ or } \begin{Vmatrix} r' & s' \\ 0 & s \end{Vmatrix}$$

where $r' = t/s$, $s'' = t/r$ and $s' = s(1-m'/m)$, $r'' = r(1-m''/m)$ for $m \triangleq \gcd(r,s) = rs/t$ and certain integers $m', m'', 0 \leqslant m', m'' < m$.[5]

We now consider the rearrangement or decimation ϕ_A of irreducible arrays, where

$$A = \begin{Vmatrix} a & b \\ c & d \end{Vmatrix},$$

$\det(A) \neq 0$. For a prime ideal $\mathfrak{P}$ belonging to the algebraic set $V(\mathfrak{P})$ which contains (α,β), let the corresponding prime ideal $\mathfrak{P}' = \psi_A^{-1}(\mathfrak{P})$ belong to the algebraic set $V(\mathfrak{P}')$. Then $V(\mathfrak{P}')$ contains (α',β'), where $\alpha' = \alpha^a\beta^b$, $\beta' = \alpha^c\beta^d$, and $K' = \dim G(\mathfrak{P}')$ is less than or equal to $K = \dim G(\mathfrak{P})$. Let $\alpha = \theta^\kappa$, $\beta = \theta^\lambda$, $\alpha' = \theta^{\kappa'}$, $\beta' = \theta^{\lambda'}$ for a fixed primitive element θ of GF (q^L). Then a rearrangement or decimation ϕ_A induces a linear transformation Φ_A from Z_N^2 into itself:

$$\kappa' \equiv a\kappa + b\lambda, \quad \lambda' \equiv c\kappa + d\lambda \bmod N\left(\triangleq q^L - 1\right) \quad (10)$$

where Z_N^2 is the direct product of the coset group $Z/(q^L-1)$ with itself. In order to give the condition that the period is left invariant under a decimation ϕ_A, we need the following lemmas.

Lemma 12: For given integers κ, λ, N, there are integers σ,τ such that $\gcd(\kappa,\lambda,N) = \gcd(\kappa+\sigma N, \lambda+\tau N)$.

Lemma 13: If $\gcd(\kappa,\lambda) = \gcd(\kappa',\lambda')$, then there exists a unimodular matrix $\boldsymbol{S}$ such that $\kappa' = a\kappa + b\lambda$, $\lambda' = c\kappa + d\lambda$, where

$$\boldsymbol{S} = \begin{Vmatrix} a & b \\ c & d \end{Vmatrix}.$$

For the proof of Lemma 13, we remark that $\mu(\triangleq \gcd(\kappa,\lambda)) = a\kappa + b\lambda$ for certain integers a,b and that, for $c \triangleq -\lambda/\mu$, $d \triangleq \kappa/\mu$, $ad - bc = 1$. The following theorem is proved in Appendix E using these lemmas.

Theorem 10: Two irreducible arrays $u \in G(\mathfrak{P})$ and $v \in G(\mathfrak{P}')$ have the same period $\text{per}\,(u) = \text{per}\,(v)$ if and only if any of the following conditions is valid:

i) there is a unimodular matrix $\boldsymbol{S}$ such that $\phi_{\boldsymbol{S}}(G(\mathfrak{P})) = G(\mathfrak{P}')$;
ii) there is a nonsingular matrix A such that $\gcd(\det(A), q^L-1) = 1$ and $\phi_A(G(\mathfrak{P})) = G(\mathfrak{P}')$;
iii) there is a pair of nonsingular matrices A and A' such that $\phi_A(G(\mathfrak{P})) = G(\mathfrak{P}')$ and $\phi_{A'}(G(\mathfrak{P}')) = G(\mathfrak{P})$.

Example 8: Let $q=3$. Consider the following irreducible polynomials over GF(3):

$$f(x) = x^2 + x + 2,$$
$$g(y) = y^4 + 2y^3 + y^2 + 2y + 1 \equiv (y^2 + xy + 1)(y^2 + 2(x+1)y + 1) \bmod f(x).$$

For a root θ of $x^4 + x + 2$, which is a primitive element of GF (3^4), $\theta^{80} = 1$, $f(x)$ has a root $\alpha = \theta^{10}$, $\alpha \in$ GF (3^2). The prime ideal $\mathfrak{P} = (x^2 + x + 2, y^2 + xy + 1)$ has the algebraic set $V(\mathfrak{P}) = \{(\theta^{10},\theta^8), (\theta^{30},\theta^{24}), (\theta^{10},\theta^{72}), (\theta^{30},\theta^{56})\}$, and $\text{per}_x\,(\mathfrak{P}) = 8$, $\text{per}_y\,(\mathfrak{P}) = 10$, $\text{per}\,(\mathfrak{P}) = 40$. For

$$\boldsymbol{S} = \begin{Vmatrix} 3 & 1 \\ 2 & 1 \end{Vmatrix}, \quad A = \begin{Vmatrix} -1 & 4 \\ 2 & -1 \end{Vmatrix},$$

$$\mathfrak{P}' \triangleq \psi_{\boldsymbol{S}}^{-1}(\mathfrak{P}) = \psi_A^{-1}(\mathfrak{P}) = (x^4 + x^3 + x^2 + 1, y + x^3 + 2x^2 + x),$$

$\text{per}_x\,(\mathfrak{P}') = 40$, $\text{per}_y\,(\mathfrak{P}') = 20$, $\text{per}\,(\mathfrak{P}') = 40$, and $V(\mathfrak{P}') = \{(\theta^{38},\theta^{28}), (\theta^{34},\theta^4), (\theta^{22},\theta^{12}), (\theta^{66},\theta^{36})\}$. An array u in $G(\mathfrak{P})$ and the rearranged array $u' = \phi_{\boldsymbol{S}}(u) \in G(\mathfrak{P}')$ are shown in Fig. 7, and the decimated array $v = \phi_A(u) \in G(\mathfrak{P}')$ is shown in Fig. 8.

Corollary 6: An arbitrary irreducible array can be rearranged to form the perpendicular repetition of a (simple) periodic sequence, that is, there is a unimodular matrix $\boldsymbol{S}$ such that $\text{per}_x\,(\phi_{\boldsymbol{S}}(u)) = \text{per}\,(u)$ and $\text{per}_y\,(\phi_{\boldsymbol{S}}(u)) = 1$.

It follows from this corollary that an irreducible two-dimensional cyclic code of code area $m \times n$ is equivalent to an irreducible cyclic code of code length mn, provided that m,n are relatively prime to q.[6]

Example 9: The 8×10 rectangle of u in Example 8 is composed of two unit rectangles. For

$$\boldsymbol{S} = \begin{Vmatrix} 1 & -1 \\ -4 & 5 \end{Vmatrix},$$

$\Phi_{\boldsymbol{S}}((10,8)) = (2,0)$, and $\mathfrak{P}' = \psi_{\boldsymbol{S}}^{-1}(\mathfrak{P}) = (x^4 + x^2 + 2x + 1, y + 1)$. An 80×1 rectangle of an array in $G(\mathfrak{P}')$ is a codeword of the cyclic code with the check polynomial $x^4 + x^2 + 2x + 1$.

The following definition[7] is a two-dimensional generalization of M-sequences [19], i.e., linear shift-register sequences with maximum length.

Definition 10: If a DP array u has $\text{per}\,(u) = q^K - 1$ for $K \triangleq \dim G(\mathfrak{A}(u))$, then it is said to be a *maximum period array* or simply an *M-array*.

The above definition is invariant under rearrangement by any unimodular transformation. If $\mathfrak{A}$ is the characteristic ideal $\mathfrak{A}(u)$ of an M-array u, $G(\mathfrak{A})$——$\{(0)\}$ consists of $q^K - 1$ translates of u which are all M-arrays with period $q^K - 1$.

The following theorem shows that an M-array is a two-dimensional analog of an M-sequence.

[5] In order to determine the integers m', m'', we remark that there exist positive integers κ', λ' relatively prime to N such that $\kappa \equiv \kappa'$ gcd (κ, N), $\lambda \equiv \lambda'$ gcd (λ, N) mod N [19]. As κ', λ' are also relatively prime to m, there exist positive integers $\kappa'^{-1}, \lambda'^{-1}$ such that $\kappa'\kappa'^{-1} \equiv \lambda'\lambda'^{-1} \equiv 1 \bmod m$. Then $m' \equiv \lambda'^{-1}\kappa'$, $m'' \equiv \kappa'^{-1}\lambda' \equiv m'^{-1} \bmod m$.

[6] In case of $\gcd(m,n) \neq 1$, the result has been unnoticed as far as the author knows.

[7] This definition differs from that of Nomura *et al.* [1], [2], where the dimension is ignored and the concept of "period" is different from ours.

Authorized licensed use limited to: Lakehead University. Downloaded on September 25,2026 at 17:14:10 UTC from IEEE Xplore. Restrictions apply.

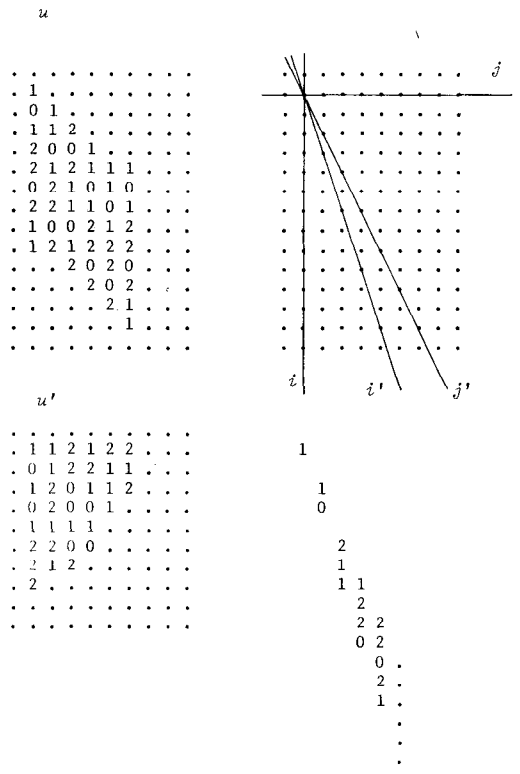


Fig. 7. Example of irreducible LR array over GF (3) and its rearrangement.

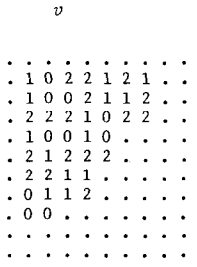


Fig. 8. Example of decimated irreducible array over GF (3).

Theorem 11: For any integral vectors $\boldsymbol{i},\boldsymbol{j}$ and arbitrary elements c,d of F, $c(_{\boldsymbol{i}}u)+d(_{\boldsymbol{j}}u)$ is either (0) or a translate of u if and only if u is an M-array.

From Definition 10, it follows that the characteristic ideal of an M-array is maximal in $\tilde{R}$. Thus we have the following lemma.

Lemma 14: The characteristic ideal $\mathfrak{A}$ of an arbitrary M-array u is prime in $\tilde{R}$, i.e., u is an irreducible array.

From this lemma, the earlier results given in this section can be also applied to M-arrays.

Lemma 15: Let $K=\dim G(\mathfrak{A}(u))$ and θ be a primitive element of GF (q^K). Provided that a zero $(\theta^{\kappa},\theta^{\lambda})$ is contained in $V(\mathfrak{A}(u))$, u is an M-array if and only if

$$\gcd(\kappa,\lambda,q^K-1)=1. \tag{11}$$

Condition (11) does not depend on the choice of a primitive element of GF (q^K) and is invariant under unimodular transformations $(\kappa',\lambda')=\Phi_S((\kappa,\lambda))$, which again implies that an M-array is rearranged to an M-array by means of any unimodular transformation.

For a primitive element θ of GF (q^K), $K=K_1K_2'$, $N\triangleq q^K-1$, $N'\triangleq q^{K_1}-1$, let $\alpha=\theta^{(N/N')\sigma'}$, $\beta=\theta^{\eta\tau'}$, where σ', τ', η are positive integers such that $\gcd(\eta,N)=1$,

$$\sigma'\tau'|N', \tag{12}$$

$$\text{if } N'/\sigma'|q^k-1, \text{ then } k\geqslant K_1, \tag{13}$$

$$\gcd(\tau',(N/N')\sigma')=1. \tag{14}$$

Nomura *et al.* [2] define a $\gamma\beta$-array to be what in our terminology is an array in $G(\mathfrak{P})$, where $\mathfrak{P}$ is the prime ideal with the algebraic set $V(\mathfrak{P})$ containing a zero (α,β) satisfying (12), (13), and (14). $\gamma\beta$-arrays are easily shown to be M-arrays. The following theorem shows that conversely all M-arrays are $\gamma\beta$-arrays. (See Appendix F.)

Theorem 12: An M-array is a $\gamma\beta$-array, and vice versa.

From Corollary 6 we can obtain a generalization of the one-to-one correspondence between a $\gamma\beta$-array and an M-sequence described in T. Nomura *et al.* [2, th. 9].

Theorem 13: An arbitrary M-array can be rearranged to the perpendicular repetition of an M-sequence with the same period.

The construction of pseudorandom arrays from pseudorandom sequences shown by F. J. MacWilliams and N. J. A. Sloane [4] is also a special case of Theorem 13.

Example 10: Let $q=2^2$, GF$(2^2)=\{0,\gamma,\gamma^2,1(=\gamma^3)\}$. For a primitive element θ of GF$(2^{2\cdot3})$ such that $\theta^3+\theta^2+\gamma\theta+\gamma^2=\theta^6+\theta+1=0$, the prime ideal $\mathfrak{P}=(x^3+\gamma x^2+1,\ y+\gamma x^2+x)$ has the algebraic set $V(\mathfrak{P})=\{(\theta^{15},\theta^{10}),(\theta^{60},\theta^{40}),(\theta^{51},\theta^{34})\}$. Thus $\text{per}_x(\mathfrak{P})=21$, $\text{per}_y(\mathfrak{P})=\text{per}(\mathfrak{P})=63$. For

$$S=\begin{Vmatrix}1 & -1\\ -2 & 3\end{Vmatrix},$$

$\mathfrak{P}'=\psi_S^{-1}(\mathfrak{P})=(x^3+\gamma x^2+\gamma^2x+\gamma,\ y+1)$, $\text{per}(\mathfrak{P}')=\text{per}_x(\mathfrak{P}')=63$, $\text{per}_y(\mathfrak{P}')=1$. A unit rectangle of an M-array $u\in G(\mathfrak{P})$ corresponds to a period of an M-sequence $v\in G(\mathfrak{P}')$ with $v_k=u_{i,j}$, where the correspondence between k and (i,j) is shown in Table I.

Theorem 13 also shows that the pseudorandomness of an M-array is relevant to that of the corresponding M-sequence. As a pseudorandom (rectangular) array, we may make use of a unit rectangle of an M-array u instead of a FP parallelogram of u. Let $\rho_u(\boldsymbol{i})$ denote the autocorrelation function of a DP array u, defined by

$$\rho_u(\boldsymbol{i})\triangleq\sum_{\boldsymbol{j}\in\Gamma}\eta(u(\boldsymbol{j}))\,\overline{\eta(u(\boldsymbol{i}+\boldsymbol{j}))},\qquad \boldsymbol{i}=(i,j)\in Z^2$$

where Γ is a unit rectangle of the array u and η is the isomorphism from the additive group of $F=\text{GF}(q)$ to the multiplicative group of complex qth roots of one. Invoking Zierler [19, corollary to th. 12], we have, in view of Theorem 13, the following result.

Authorized licensed use limited to: Lakehead University. Downloaded on September 25,2026 at 17:14:10 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CONSTRUCTION OF A UNIT RECTANGLE OF AN M-ARRAY FROM AN M-SEQUENCE

i \ j	0	1	2
0	0	2	4
1	3	5	7
2	6	8	10
3	9	11	13
4	12	14	16
5	15	17	19
6	18	20	22
7	21	23	25
8	24	26	28
9	27	29	31
10	30	32	34
11	33	35	37
12	36	38	40
13	39	41	43
14	42	44	46
15	45	47	49
16	48	50	52
17	51	53	55
18	54	56	58
19	57	59	61
20	60	62	1

Theorem 14: Let $t=(q^K-1)/(q-1)$. An M-array u with period q^K-1 has the following autocorrelation function:

i) for $\boldsymbol{i}=(i,j)$ such that $ai+cj \not\equiv tk \bmod q^K-1$, $0 \leqslant k \leqslant q^K-1$, $\rho(\boldsymbol{i})=q^{K-2}\Sigma_{\lambda,\mu\in F}\eta(\lambda)\overline{\eta(\mu)}-|\eta(0)|^2$,
ii) for $\boldsymbol{i}=(i,j)$ such that $ai+cj \equiv tk \bmod q^K-1$, $0 \leqslant k \leqslant q^K-1$, $\rho(\boldsymbol{i})=q^{K-1}\Sigma_{\lambda\in F}\eta(\lambda)\overline{\eta(\nu^k\lambda)}-|\eta(0)|^2$,

where

$$\boldsymbol{S}=\begin{Vmatrix} a & b \\ c & d \end{Vmatrix}$$

is the unimodular matrix corresponding to the rearrangement from the perpendicular repetition of an M-sequence to u and ν is a primitive element of F.

In particular, for $q=2$, we have the following corollary.

Corollary 7: An M-array over GF (2) with period $t=2^K-1$ has the autocorrelation function

$$\rho(\boldsymbol{i})=\begin{cases} -1, & \boldsymbol{i}\neq m\boldsymbol{l}_1+n\boldsymbol{l}_2 \\ t, & \boldsymbol{i}=m\boldsymbol{l}_1+n\boldsymbol{l}_2 \end{cases}$$

where $(m,n)\in Z^2$ and $\boldsymbol{l}_1\times\boldsymbol{l}_2$ is a FP parallelogram of u.

When applied to M-arrays, Theorem 10 yields the following theorem.

Theorem 15: Two M-arrays u,v have the same period $t=q^K-1$ if and only if there exists a unimodular transformation $\boldsymbol{S}$ such that $v=\phi_{\boldsymbol{S}}(\boldsymbol{l}u)$ for certain $\boldsymbol{l}\in Z^2$.

VI. CONCLUSION

We have given a general treatment of doubly periodic arrays over a finite field GF (q). Linear recurring arrays over GF (q) are shown to coincide with DP arrays. A DP array is determined by a system of LR relations, given the values in an independent point set. Any independent point set associated with the system of LR relations can be obtained in general.

Those properties which are invariant under rearrangements of arrays by unimodular transformations are the most important in studying DP arrays. In particular, rearrangement, decimation, and interleaving determine correspondences between arrays.

The general treatment of DP arrays is applied to irreducible arrays. M-arrays, which are a special class of irreducible arrays, are dealt with from the same point of view. A two-dimensionally proper definition of M-array is given.

Correspondences between irreducible arrays are clarified by means of rearrangements and decimations. The general construction of M-arrays from M-sequences is given by unimodular transformations. M-arrays are shown to coincide with $\gamma\beta$-arrays. Thus it becomes clear that there exist no arrays having "maximum period" in our terminology other than $\gamma\beta$-arrays.

DP arrays are a natural generalization of linear recurring sequences [19]. Using our results, we can implement LR arrays and their transformations by a two-dimensional feedback shift register. It is also easy to extend our investigation to multidimensional arrays.

ACKNOWLEDGMENT

The author wishes to express his sincere thanks to Prof. M. Iri of the University of Tokyo for his valuable advice. The idea of "rearrangement" was suggested by him. Particular thanks are due to Dr. T. S. Han of the Sagami Institute of Technology, who read parts of the manuscript and contributed many helpful discussions. The author is also indebted to Dr. H. Imai of Yokohama National University for his very helpful comments.

APPENDIX A
"THE RING $\tilde{R}$ OF OPERATORS IS NOETHERIAN"

For an ideal $\tilde{\mathfrak{A}}$ in $\tilde{R}$, $\mathfrak{A} \triangleq \tilde{\mathfrak{A}} \cap R$ is an ideal in the polynomial ring R. Conversely, the ideal $\tilde{\mathfrak{A}}$ in $\tilde{R}$ is generated by the elements of $\mathfrak{A}$. Thus an ideal in $\tilde{R}$ is in one-to-one correspondence with an ideal in R, from which it follows that the divisor chain condition [17] is valid in $\tilde{R}$ as well as in R. This means that $\tilde{R}$ is also a Noetherian ring and almost all theorems about ideals in R can be restated for ideals in $\tilde{R}$. Every ideal in $\tilde{R}$ has a finite basis $(f^{(1)},\cdots,f^{(L)})$, where $f^{(1)},\cdots,f^{(L)}$ may be chosen to be polynomials. Furthermore, the Lasker–Noether decomposition theorem is also valid [17], [18]: every ideal $\mathfrak{A}$ has an irredundant representation as the intersection of a finite number of greatest primary components, $\mathfrak{A}=\cap_i\mathfrak{A}_i$. These primary ideals $\mathfrak{A}_i$ belong to distinct prime ideals $\mathfrak{P}_i$.

Authorized licensed use limited to: Lakehead University. Downloaded on September 25,2026 at 17:14:10 UTC from IEEE Xplore. Restrictions apply.

APPENDIX B
SOME THEOREMS ON R-ISOMORPHISMS [18, PP. 132–144]

Let R be a commutative ring. For submodules N and L of an R-module M with $N \subset L$, the *difference module* $L-N$ (or the factor module L/N) is a submodule of $M-N$, and $(M-N)-(L-N) \cong M-L$, where $M \cong M'$ indicates that M and M' are R-isomorphic.

If N and L are submodules of an R-module M, then $(L+N)-N \cong L-(L \cap N)$, where $L+N$ is the "sum" of L, N.

Let $\mathfrak{A}$ be an ideal in a ring R. The residue class ring $R/\mathfrak{A}$ and the difference module $R-\mathfrak{A}$ are identical as sets, as additive groups (modules), and as $R/\mathfrak{A}$-modules.

If R contains the finite field F, an R-module is also an F-module. A finite R-module M has finite dimension $\dim_F(M)$ as an F-module. From the above isomorphism theorems, we have the following lemmas.

Lemma B1: For ideals $N, \mathfrak{A}$ in R with $N \subset \mathfrak{A}$, $\dim_F(R/N) - \dim_F(\mathfrak{A}/N) = \dim_F(R/\mathfrak{A})$.

Lemma B2: For ideals $\mathfrak{A}, \mathfrak{B}$ in R, $\dim_F(\mathfrak{A}+\mathfrak{B}/\mathfrak{B}) = \dim_F(\mathfrak{A}/\mathfrak{A} \cap \mathfrak{B})$.

APPENDIX C
DETERMINATION OF THE CHARACTERISTIC IDEAL $\mathfrak{A}(u)$

We begin by finding an independent point set $\Delta \triangleq \Delta(\mathfrak{A}(u))$ of the characteristic ideal $\mathfrak{A}(u)$ of u. For any finite subset Γ of Z^2, the *periphery* $\partial\Gamma$ of Γ is defined as: $\partial\Gamma = \{\boldsymbol{l}+\boldsymbol{i}_x | \boldsymbol{l} \in \Gamma\} \cup \{\boldsymbol{l}+\boldsymbol{i}_y | \boldsymbol{l} \in \Gamma\} - \Gamma$. Let Γ be a FP parallelogram (or a unit rectangle) of u and $\Gamma_1 = \{\boldsymbol{0}\}$. For $k = 1, 2, \cdots$, $\Gamma_{k+1} \triangleq \Gamma_k + \{\boldsymbol{m}_k\}$, where $\boldsymbol{m}_k \in \partial\Gamma_k$ is chosen so that, for a certain $\boldsymbol{i} \in \Gamma$ and for arbitrary $f_j \in F$, $\boldsymbol{j} \in \Gamma_k$, $u_{i+m_k} + \sum_{j \in \Gamma_k} f_j u_{i+j} \neq 0$. If there exists no such $\boldsymbol{m}_k \in \partial\Gamma_k$ for some k, then $\Delta = \Gamma_k$ and $k = \dim G(\mathfrak{A}(u))$.

For each $\boldsymbol{l} = (k,l) \in \partial\Delta$, we find a polynomial $f_l(x,y) = x^k y^l + \sum_{(i,j) \in \Delta} f_{i,j} x^i y^j$ satisfying $f_l u = (0)$. Then $\mathfrak{A} = (\{f_l | \boldsymbol{l} \in \partial\Delta\})$ is the characteristic ideal $\mathfrak{A}(u)$.

APPENDIX D
PROOF OF THEOREM 8

From (7), $\phi_A^*(G(\mathfrak{A})) = \phi_A^{-1}(G(\mathfrak{A}))$. In view of the linearity of ϕ_A, for $u, v \in \phi_A^{-1}(G(\mathfrak{A}))$ and $c, d \in F$, we have $\phi_A(cu+dv) = c\phi_A(u) + d\phi_A(v) \in G(\mathfrak{A})$, from which it follows that $cu+dv \in \phi_A^{-1}(G(\mathfrak{A}))$. From the definition of ϕ_A^*, $xu \in \phi_A^*(G(\mathfrak{A}))$, provided that $u \in \phi_A^*(G(\mathfrak{A}))$. Let $r = \mathrm{per}_x(\mathfrak{A})$, $s = \mathrm{per}_y(\mathfrak{A})$. Then for $u \in \phi_A^*(G(\mathfrak{A}))$, we have $u_{i+ark+csl, j+brk+dsl} = u_{i,j}$, $(i,j), (k,l) \in Z^2$. In particular, $u_{i+rs(ad-bc), j} = u_{i,j} = u_{i,j+rs(ad-bc)}$, $(i,j) \in Z^2$. Thus $\phi_A^*(G(\mathfrak{A}))$ is an $\tilde{R}$-module of DP arrays, which implies, in view of Theorem 1, that there is an ideal $\mathfrak{A}_A$ in $\tilde{R}$ such that $\phi_A^*(G(\mathfrak{A})) = G(\mathfrak{A}_A)$. Since $\phi_A(G(\mathfrak{A}_A)) = G(\psi_A^{-1}(\mathfrak{A}_A)) = G(\mathfrak{A})$, $\psi_A^{-1}(\mathfrak{A}_A) = \mathfrak{A}$ by virtue of Lemma 6. Furthermore, an arbitrary array $u \in \phi_A^*(G(\mathfrak{A}))$ satisfies, for $f(x,y) \in \mathfrak{A}$, $\sum_{k \in \Gamma(f)} f_k u(\boldsymbol{(k+i)A+m}) = \sum_{k \in \Gamma(f)} f_k v(\boldsymbol{k+i}) = 0$, where $v = \phi_A(_m u) \in G(\mathfrak{A})$, $\boldsymbol{i}, \boldsymbol{m} \in Z^2$. Therefore, $\sum_{k \in \Gamma(f)} f_k u(\boldsymbol{kA+i}) = 0$, $\boldsymbol{i} \in Z^2$, from which it follows that $f'(x,y)u = (0)$, where $f'(x,y) = \psi_A(f(x,y)) \in \psi_A(\mathfrak{A})$. Accordingly $\mathfrak{A}_A$ contains $\psi_A(\mathfrak{A})$. Suppose $\mathfrak{B}$ be any ideal containing $\psi_A(\mathfrak{A})$. Then $\psi_A^{-1}(\mathfrak{B}) \supset \psi_A^{-1}(\psi_A(\mathfrak{A})) = \mathfrak{A} + \ker \psi_A = \mathfrak{A} = \psi_A^{-1}(\mathfrak{A}_A)$, where the kernel $\ker \psi_A$ of the one-to-one map ψ_A is the null ideal (0). Therefore, $\mathfrak{B} \supset \mathfrak{A}_A$.

The final identity follows from the fact that $|Z^2/Z_A^2|$ is equal to the absolute value of $\det(\boldsymbol{A})$ and that there is a unique array u corresponding to choosing the arrays in $G(\mathfrak{A})$ one by one from each coset. Q.E.D.

APPENDIX E
PROOF OF THEOREM 10

Condition i) is a special case of ii), and ii) is a special case of iii), because in the case of ii) there exists an inverse map Φ_A^{-1}. Under condition iii), there are integers e, f such that $\kappa' = a\kappa + b\lambda + e(q^L-1)$, $\lambda' = c\kappa + d\lambda + f(q^L-1)$, from which it follows that $\gcd(\kappa, \lambda, q^L-1) | \kappa', \lambda'$. Then we have $\gcd(\kappa, \lambda, q^L-1) | \gcd(\kappa', \lambda', q^L-1)$. Similarly, $\gcd(\kappa', \lambda', q^L-1) | \gcd(\kappa, \lambda, q^L-1)$, so that we have $\gcd(\kappa, \lambda, q^L-1) = \gcd(\kappa', \lambda', q^L-1)$. Consequently $\mathrm{per}(u) = \mathrm{per}(v)$. Conversely, assuming $\gcd(\kappa, \lambda, q^L-1) = \gcd(\kappa', \lambda', q^L-1)$, we have, by virtue of Lemma 12, $\gcd(\kappa + \sigma(q^L-1), \lambda + \tau(q^L-1)) = \gcd(\kappa' + \sigma'(q^L-1), \lambda' + \tau'(q^L-1))$ for certain integers $\sigma, \tau, \sigma', \tau'$. Therefore, in view of Lemma 13, there exists a unimodular matrix

$$\boldsymbol{S} = \begin{Vmatrix} a & b \\ c & d \end{Vmatrix}$$

such that $(\kappa', \lambda') = \Phi_{\boldsymbol{S}}((\kappa, \lambda))$. Q.E.D.

APPENDIX F
PROOF OF THEOREM 12

Let $N' = q^{K_1} - 1$, $N'' = q^{K_2} - 1$. For a primitive element $\theta_1 = \theta^{N/N'}$ of $\mathrm{GF}(q^{K_1})$ and a primitive element $\theta_2 = \theta^{N/N''}$ of GF (q^{K_2}), $\alpha = \theta^{\kappa}$, $\beta = \theta^{\lambda}$ can be rewritten as $\alpha = \theta_1^{\sigma}$, $\beta = \theta_2^{\tau}$. Therefore, $\kappa = (N/N')\sigma$, $\lambda = (N/N'')\tau$. Condition (11) implies that $\gcd(N/N', N/N'') = 1$, from which it follows that $N = \mathrm{lcm}(N', N'')$. This yields $N \leqslant N'N'' < q^{K_1+K_2} - 1$ and thus $K < K_1 + K_2$. Then, assuming that $K_2 \geqslant K_1$, $K_2 \geqslant K_1 > K - K_2 = K_2(K_1' - 1)$, and $K_1' = 1$, i.e., $K = K_2$. In this case $\lambda = \tau$. Let $\sigma \equiv \xi\sigma' \bmod N'$, $\tau \equiv \zeta\tau' \bmod N$, where $\sigma' \triangleq \gcd(\sigma, N')$, $\tau' \triangleq \gcd(\tau, N)$, and $\gcd(\xi, N') = \gcd(\zeta, N) = 1$ [19]. $\theta_1' \triangleq \theta_1^{\xi}$ is a primitive element of GF (q^{K_1}) and $\theta_2' \triangleq \theta^{\zeta}$ is a primitive element of GF (q^K). There exists a primitive element θ' of GF (q^K) such that $\theta_1' = \theta'^{N/N'}$, $\theta_2' = \theta'^{\eta}$, for a certain integer η relatively prime to N. Thus, writing θ instead of θ', we can put $\kappa = \sigma'(N/N')$, $\lambda = \eta\tau'$. Condition (13) is clear. Condition (11) implies that $\gcd(\sigma', \tau', N)$ $(= \gcd(\sigma', \tau')) = 1$ and $\gcd(N/N', \tau') = 1$, from which it follows that $1 = \gcd(N/N', \lambda, N) = \gcd(\lambda, N/N')$, i.e., $\tau' = \gcd(\lambda, N') | N'$. Therefore $\sigma'\tau' | N'$ (12). $\gcd(\tau', \sigma'(N/N')) = 1$ (14) has already been established. Q.E.D.

REFERENCES

[1] T. Nomura and A. Fukuda, "Linear recurring planes and two-dimensional cyclic codes" (in Japanese), *IECE Trans.*, vol. 54-A, pp. 147–154, Mar. 1971.

[2] T. Nomura, H. Miyakawa, H. Imai, and A. Fukuda, "A theory of two-dimensional linear recurring arrays," *IEEE Trans. Inform. Theory*, vol. IT-18, pp. 775–785, Nov. 1972.

[3] T. Ikai, H. Kosako, and Y. Kojima, "Two-dimensional linear recurring arrays—Their formulation by ideals and eventually periodic arrays," *IECE Trans.*, vol. 60-A, pp. 123–130, Feb. 1977.

[4] F. J. MacWilliams and N. J. A. Sloane, "Pseudo-random sequences and arrays," *Proc. IEEE*, vol. 64, pp. 1715–1729, 1976.

[5] S. Sakata, "Multi-fold linear recurring arrays and multi-fold M-arrays" (in Japanese), Technical Group on Automata and Languages, IECE, Japan, AL76-66, Dec. 1976.

[6] ——, "Multi-fold linear recurring arrays and their rearrangements" (in Japanese), Technical Group on Automata and Languages, IECE, Japan, AL76-84, Mar. 1977.

[7] ——, "Doubly linear recurring arrays and M-arrays," *IECE Trans.*, vol. 60-A, pp. 918–925, Oct. 1977.

[8] H. Hancock, *Lectures on the Theory of Elliptic Functions*, vol. 1, *Analysis*. New York: Wiley, 1910.

Authorized licensed use limited to: Lakehead University. Downloaded on September 25,2026 at 17:14:10 UTC from IEEE Xplore. Restrictions apply.

[9] S. D. Berman, "Semi-simple cyclic and Abelian codes," *Cybernetics*, vol. 3, no. 1, pp. 21–30, 1967.
[10] P. Delsarte, "Automorphisms of Abelian codes," *Philips Res. Rep.*, vol. 25, pp. 389–403, 1970.
[11] F. J. MacWilliams, "Binary codes which are ideals in the group algebra of an Abelian group," *Bell Syst. Tech. J.*, vol. 49, pp. 987–1011, 1970.
[12] H. Imal, "Two-dimensional fire codes," *IEEE Trans. Inform. Theory*, vol. IT-19, pp. 796–806, Nov. 1973.
[13] M. Arakaki and H. Imai, "Theory of two-dimensional cyclic codes of even area" (in Japanese), Technical group on Automata and Languages, IECE, Japan, AL73-77, Feb. 1974.
[14] T. Ikai, H. Kosako, and Y. Kojima, "Basic theory of two-dimensional cyclic codes—Periods of ideals and fundamental theorems," *IECE Trans.*, vol. 59-A, pp. 216–223, Mar. 1976.
[15] ——, "Basic theory of two-dimensional cyclic codes—Structure of cyclic codes and their dual codes," *IECE Trans.*, vol. 59-A, pp. 224–231, Mar. 1976.
[16] ——, "Basic theory of two-dimensional cyclic codes—Generator polynomials and the positions of check symbols," *IECE Trans.*, vol. 59-A, pp. 311–318, Apr. 1976.
[17] B. L. van der Waerden, *Modern Algebra*, vol. II (F. Blum, English transl.). New York: Ungar, 1949.
[18] O. Zariski and P. Samuel, *Commutative Algebra*, vol. I. Toronto: van Nostrand, 1958.
[19] N. Zierler, "Linear recurring sequences," *J. Soc. Indust. Appl. Math.*, vol. 7, pp. 31–48, 1959.

Generalized Quadratic Residue Codes

JACOBUS H. VAN LINT AND F. JESSIE MACWILLIAMS

***Abstract*—A simple definition of generalized quadratic residue codes, that is, quadratic residue codes of block length p^m, is given, and an account of many of their properties is presented.**

I. INTRODUCTION

LET p,l be distinct primes such that l is a quadratic residue of p. Let $\mathbb{F}=\mathrm{GF}(l)$, and let G be the Abelian group of the additive structure of $\mathrm{GF}(p)$. The (classical) quadratic residue codes A^+,B^+,A,B are certain ideals in the group algebra $\mathbb{F}G$. G is of course a cyclic group, and the group operation is written as multiplication, i.e., $G=\{1,x,x^2,\cdots,x^{p-1}\}$. A^+ is defined as follows. Let ξ be a primitive pth root of unity over $\mathbb{F}$; a polynomial $c(x)$ of $\mathbb{F}G$ is in A^+ if $c(\xi^r)=0$ for all r which are quadratic residues of p. The code A has one as an additional zero. B^+,B are defined similarly with respect to the nonresidues of p. (See [1], [6], [8].)

We wish to extend this idea to codes of block length $q=p^m$, $m>1$. These will be called generalized quadratic residue (GQR) codes. The restrictions on $\mathbb{F}$ will be described later; in fact, if m is even, there are no restrictions except that $\mathbb{F}$ should not have characteristic p. $\mathbb{F}$ can even be taken to be the real numbers. G is now the Abelian group of the additive structure of $\mathrm{GF}(p^m)$. It is no longer cyclic, but is an elementary Abelian group of type $\{p,p,\overset{\leftarrow m\rightarrow}{\cdots},p\}$, which means that it is the direct product of m cyclic groups of order p. The code positions will be identified with the elements of G. Since we need addition and multiplication both in $\mathrm{GF}(p^m)$ and in the group algebra $\mathbb{F}G$, we use the symbols $\oplus$ and $*$ for these operations in $\mathbb{F}G$; a sum in $\mathbb{F}G$ will be denoted by $\sum\!\!\!\!\!\!\sum$.

We remind the reader that $\mathbb{F}G$ consists of all formal sums $\sum\!\!\!\!\!\!\sum_{g\in G}a_g g$, $a_g\in\mathbb{F}$, with the following rules for addition and multiplication:

$$\Big(\sum\!\!\!\!\!\!\sum_{g\in G}a_g g\Big)\oplus\Big(\sum\!\!\!\!\!\!\sum_{g\in G}b_g g\Big)=\sum\!\!\!\!\!\!\sum_{g\in G}(a_g+b_g)g \tag{1}$$

$$\Big(\sum\!\!\!\!\!\!\sum_{g\in G}a_g g\Big)*\Big(\sum\!\!\!\!\!\!\sum_{g\in G}b_g g\Big)=\sum\!\!\!\!\!\!\sum_{g\in G}\Big(\sum_{g_1+g_2=g}a_{g_1}b_{g_2}\Big)g. \tag{2}$$

A subset S of G can be interpreted as an element of $\mathbb{F}G$ by taking $a_g=1$ if $g\in S$ and $a_g=0$ if $g\notin S$. We use the same symbol for the set and the corresponding element of $\mathbb{F}G$. In particular, $U,V,0$ are the elements of $\mathbb{F}G$ corresponding, respectively, to the set of nonzero squares of $\mathrm{GF}(q)$, the set of nonsquares, and the single element $\{0\}$. A vector $\boldsymbol{c}$ with coordinates c_g is identified with the element $\sum\!\!\!\!\!\!\sum_{g\in G}c_g g$ of $\mathbb{F}G$. We denote the usual inner product of two vectors $\boldsymbol{c},\boldsymbol{c}'$ by $\langle \boldsymbol{c}\cdot\boldsymbol{c}'\rangle$.

A character ψ of G is a homormorphism of G into the set of pth roots of unity over $\mathbb{F}$. The characters of G form a group χ which is isomorphic in many ways to G, let us say $\psi_g\leftrightarrow g$. The exact form of the isomorphism we need will be discussed in Section II. For a detailed account of the properties of characters as applied to coding theory, see [7]. A character is extended in the obvious way to a linear functional on the group algebra

$$\psi_f\Big(\sum\!\!\!\!\!\!\sum a_g g\Big)=\sum \alpha_g\psi_f(g).$$

Manuscript received February 9, 1978.
J. H. van Lint was with Bell Laboratories, Murray Hill, NJ, on leave from the Department of Mathematics, Technological University Eindhoven, Eindhoven, The Netherlands.
F. J. MacWilliams is with Bell Laboratories, Murray Hill, NJ 07974.

0018-9448/78/1100-0730$00.75 ©1978 IEEE

Authorized licensed use limited to: Lakehead University. Downloaded on September 25,2026 at 17:14:10 UTC from IEEE Xplore. Restrictions apply.